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Anonymous replication dossier: correlated displacement regression

Scientific objective

Reproduce a complete stochastic experiment comparing ordinary, weighted, and generalized least-squares estimates of a diffusion coefficient from correlated mean-squared-displacement (MSD) observations. Generate every trajectory from the prescribed seeds; do not substitute a published MSD curve or sample from a fitted distribution.

Random-walk experiment

For each integer seed from 0 through 4095, initialize an independent legacy MT19937 random-number generator using the semantics of ‘numpy.random.RandomState(seed)‘. Simulate 128 particles, each taking 128 steps on a three-dimensional cubic lattice. Draw all direction indices in one call equivalent to ‘rng.choice(6, size=(128,128))‘: rows index particles and columns index consecutive timesteps. Map choices, in order, to displacement vectors

[EQUATION]

    (+j,0,0), (-j,0,0), (0,+j,0), (0,-j,0), (0,0,+j), (0,0,-j),
    where j = 2.4494897428.

This decimal step length is part of the fixed workflow and makes the true diffusion coefficient numerically equal to one in the dimensionless timestep convention because <|Delta r|^2> = 6 D Delta t.

For a lag k in {1,...,128}, include the implicit position at time zero and calculate all valid time-origin displacements for each particle:

[EQUATION]

    MSD_s(k) = 1 / [128(129-k)]
        sum_{p=1}^{128} sum_{t=0}^{128-k}
        |r_{s,p}(t+k)-r_{s,p}(t)|^2.

The submitted ‘msd‘ array contains lags k=2,...,128 in increasing order and has one row per seed, producing shape (4096,127). Lag one is excluded from the regression experiment.

Covariance and regressions

Compute the unbiased sample covariance across the 4096 MSD curves, using the 4095 denominator. Submit this 127 by 127 matrix as ‘covariance‘. Let the time vector be t=(1,2,...,127), matching the historical index convention for the selected MSD columns, and use a design matrix with columns [t,1]. Thus every fit includes an intercept. For a response vector y, calculate

[EQUATION]

    beta_hat = (A^T W A)^(-1) A^T W y,
    D_hat = beta_hat[0] / 6.

Use W=I for OLS, ‘numpy.linalg.pinv‘ with ‘rcond=1e-15‘ on the diagonal covariance for WLS, and the same pseudoinverse convention on the full covariance for GLS. Apply each fixed weight matrix to all 4096 response curves. Submit ‘diffusion_estimates‘ with rows ordered OLS, WLS, GLS and shape (3,4096).

Summary statistics

Submit a JSON object named ‘summary‘, keyed exactly by ‘OLS‘, ‘WLS‘, and ‘GLS‘. For every method provide ‘population_mean‘ and ‘population_standard_deviation‘; the latter uses the population convention with denominator 4096. Also provide ‘analytical_uncertainty‘. For WLS and GLS, this is sqrt([(A^T W A)^(-1)]_00)/6. For OLS, calculate the conventional intercept-inclusive slope standard error separately for each response curve, divide each by six, and report their arithmetic mean.

All arrays must be finite, numeric, non-pickle NPY files. The covariance must be symmetric and positive semidefinite up to ordinary floating-point roundoff.

[MASKED: all generated arrays, estimator means, estimator spreads, analytical uncertainties, and numerical comparisons are withheld.]